

Same card. Two very different shadows.

Learning intention

Describe how moving an opaque card changes its shadow size when the source and screen stay fixed.

The card stays 4 cm high.

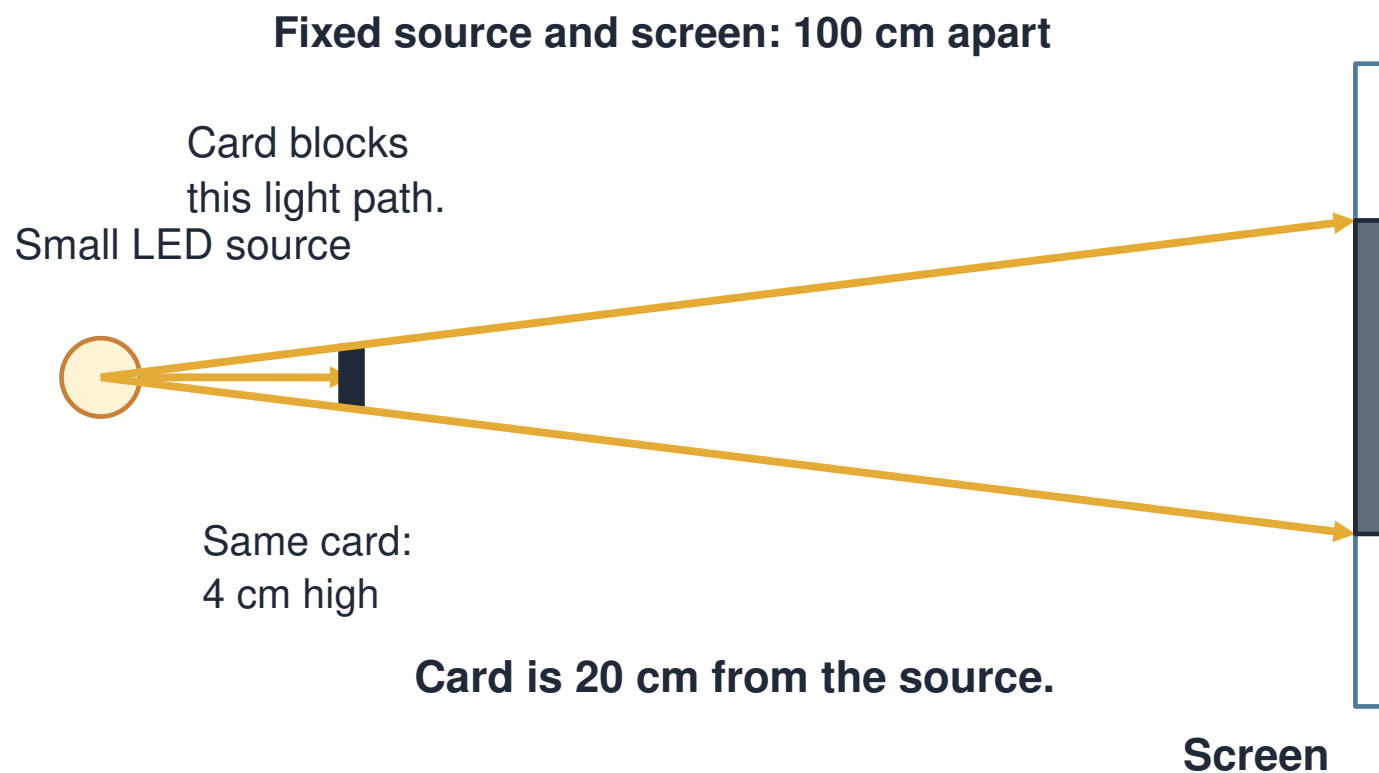
The source and screen stay still.

Can its shadow change size?

Keep the shadow idea clear

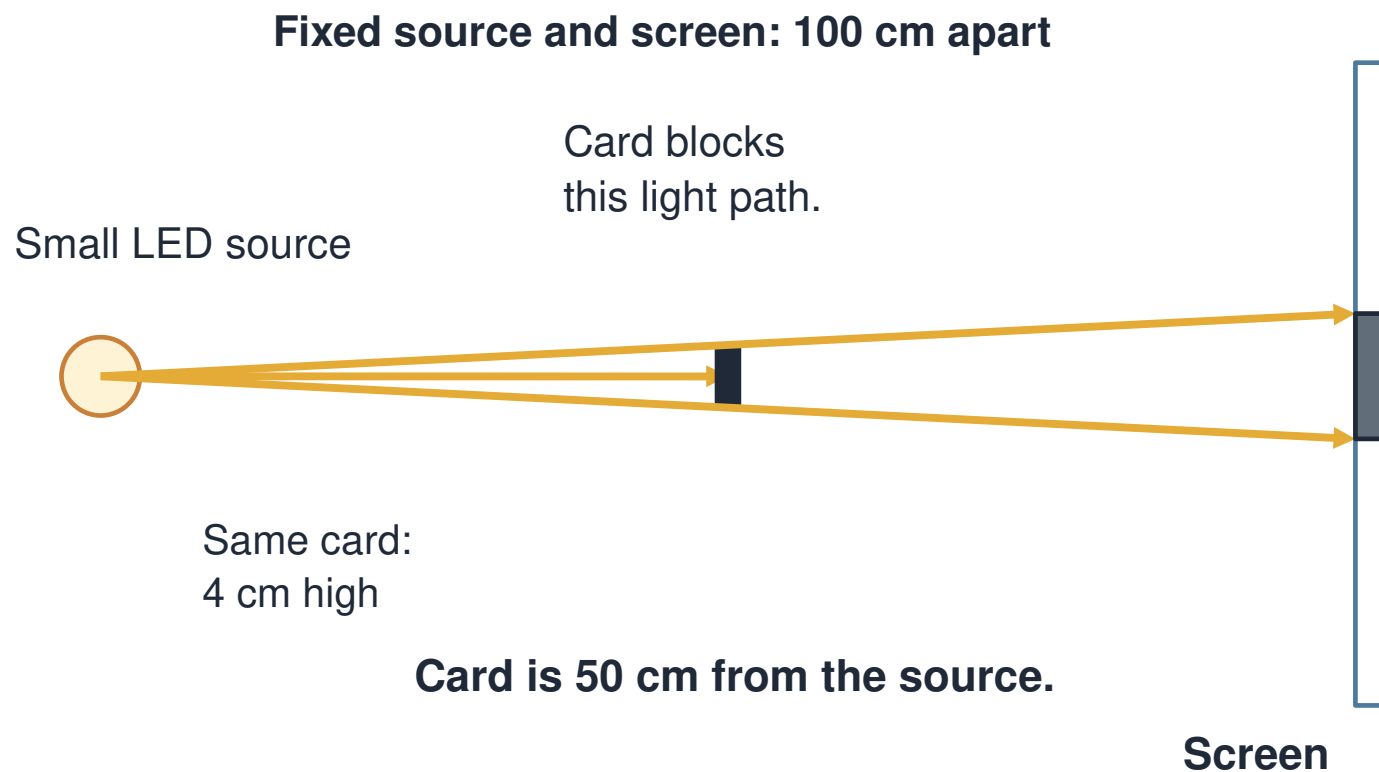
1. What does an opaque card do?
2. Where is the shadow seen?
3. Is the shadow a solid object?

I move the card near the source



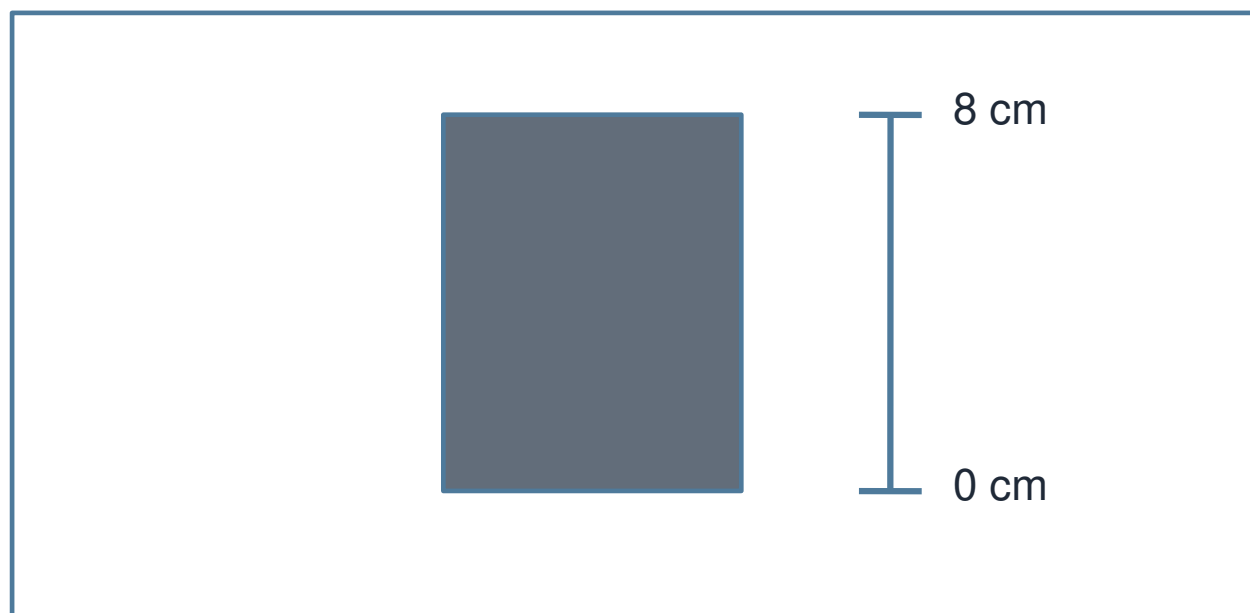
The source and screen stay fixed. The card is 20 cm from the source. The same card blocks a wider spread.

I move it towards the screen



Now the card is 50 cm from the source. Its size is unchanged. The shadow on the screen is smaller.

I measure the same dimension



Measure from the bottom edge to the top edge.

Measure bottom to top. Use centimetres each time. Keep the ruler beside the screen.

Find the pattern in the model

Compare the three positions. Which gives the largest shadow? Use two numbers as evidence.

Source → card	Card height	Model shadow height
20 cm	4 cm	20 cm
40 cm	4 cm	10 cm
50 cm	4 cm	8 cm

Constructed ideal point-source model; source → screen = 100 cm

W1. What changes as the card moves away from the source?

Meet the studio target

Target: a shadow 10 cm high.

Choose a card position.

Explain why the other positions miss.

W2. Which model position meets the target?

Which prediction follows this setup?

The source and screen stay fixed. Move the same card towards the source.

- A** The shadow becomes larger.
- B** The card itself becomes taller.
- C** The shadow must stay the same height.

Choose your investigation route

Match the model values.

Recommend a studio position.

Predict between trials.

Use the evidence

Read or point to one clue.

Record what it supports.

Keep your explanation short.

Make your own science record

Complete your chosen route.

Use a diagram, words or speech.

Check one answer with the evidence.

Check and improve

Does my answer match the evidence?

Have I explained why?

Improve one label or sentence.

Lundy loop: improve our account

Choose how to share one idea.

Listen to another explanation.

Use evidence to improve our account.

What must we name when we say “nearer makes it larger”?

Three quick checks

- E1. What happens to shadow size as this card moves towards the source?
- E2. Name one thing kept fixed.
- E3. Which model distance gives a 10 cm shadow?